

# Voice Based Concept Understanding Analyser

## Project Description:

The Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered educational assessment platform designed to evaluate a student's conceptual understanding through spoken explanations. The application combines speech-to-text transcription, semantic similarity analysis, audio signal processing, and AI-based scoring to assess both conceptual knowledge and communication effectiveness.

Built using Python and Streamlit, the platform integrates OpenAI Whisper for speech recognition, Sentence-BERT for semantic similarity, Librosa for audio feature extraction, Matplotlib for waveform visualization, and ReportLab for automated PDF report generation. The system provides an interactive dashboard that displays transcription, waveform visualization, semantic similarity score, filler word analysis, pause ratio, confidence level, final understanding score, and personalized feedback.

The application serves as an intelligent educational tool for students, educators, trainers, and researchers, helping users improve conceptual knowledge, communication skills, and presentation confidence.

## Scenarios

### Scenario 1: Semantic Understanding Evaluation

A student uploads an audio explanation of a concept such as Machine Learning, Cloud Computing, or Artificial Intelligence. The system transcribes the speech using OpenAI Whisper and compares the explanation with a predefined reference concept using Sentence-BERT semantic embeddings. Based on semantic similarity, the application evaluates conceptual understanding and classifies the result as Strong Understanding, Moderate Understanding, or Poor Understanding.

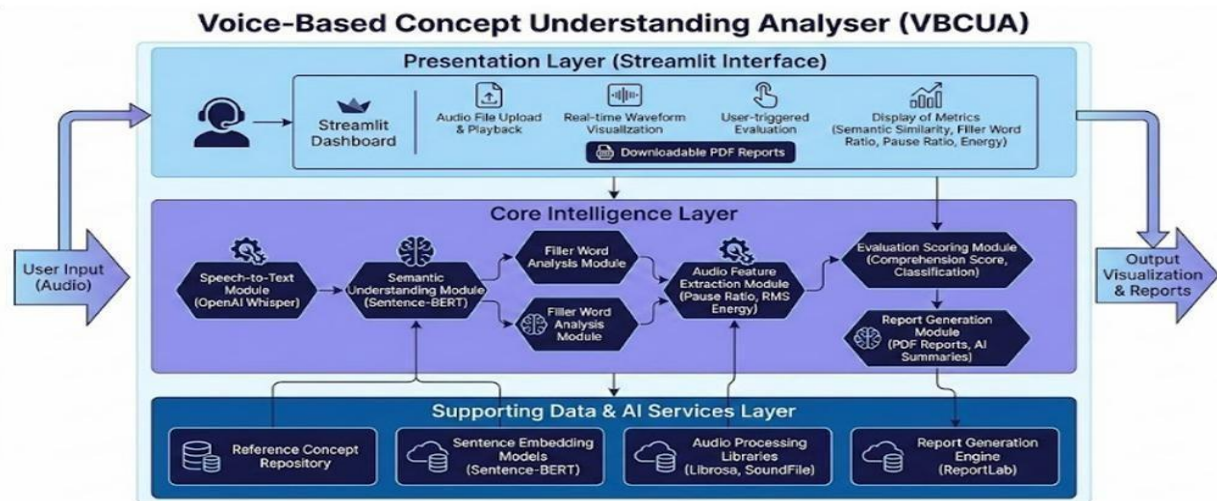
### Scenario 2: Speech Fluency and Communication Analysis

A learner preparing for interviews or presentations records a spoken explanation. The application analyzes speech fluency using filler word detection, pause ratio analysis, RMS energy, and audio signal processing through Librosa. The generated feedback helps users improve confidence, articulation, speaking clarity, and overall communication skills.

### Scenario 3: Interactive Performance Reporting

After analysis, the application displays an interactive dashboard containing the transcribed explanation, waveform visualization, semantic similarity score, communication metrics, confidence level, and final evaluation score. Users can download a professionally formatted PDF report containing detailed analysis and AI-generated feedback for academic assessment and future review.

## Technical Architecture:



**Description:** The Voice-Based Concept Understanding Analyser follows a modular AI-driven architecture that integrates speech recognition, semantic analysis, audio processing, scoring, and reporting into a single educational assessment platform.

The Streamlit frontend allows users to upload voice recordings and visualize evaluation results. The backend uses OpenAI Whisper for speech transcription, Sentence-BERT for semantic similarity computation, Librosa for extracting speech features, and Python-based scoring algorithms to evaluate conceptual understanding. Matplotlib generates waveform visualizations, while ReportLab creates downloadable PDF reports. The modular architecture ensures scalability, maintainability, and efficient AI-based evaluation.

## Pre-requisites:

**Python 3.10+ :** <https://docs.python.org/3/>

**Streamlit**<https://docs.streamlit.io/>

**OpenAI Whisper**<https://github.com/openai/whisper>

**Sentence-Transformers (Sentence-BERT) :** <https://www.sbert.net/>

**PyTorch (Torch)**<https://pytorch.org/docs/stable/index.html>

**Librosa**<https://librosa.org/doc/latest/index.html>

**NumPy**<https://numpy.org/doc/>

**Matplotlib**<https://matplotlib.org/stable/users/index.html>

**ReportLab**<https://www.reportlab.com/documentation/>

**SoundFile**<https://python-soundfile.readthedocs.io/>

**Git**[https : //git-scm.com/doc](https://git-scm.com/doc)

**GitHub**[https : //docs.github.com/](https://docs.github.com/)

**Visual Studio Code**[https : //code.visualstudio.com/docs](https://code.visualstudio.com/docs)

## **Project Workflow:**

### **Milestone 1: Model Selection and Architecture**

- **Activity 1.1:** Install and Configure Required Libraries

- **Activity 1.2:** Research and Select the Appropriate AI Models
- **Activity 1.3:** Define the Architecture of the Application
- **Activity 1.4:** Set Up the Development Environment

### **Milestone 2: Core Functionalities Development**

- **Activity 2.1:** Develop Speech-to-Text Transcription Module
- **Activity 2.2:** Implement Semantic Similarity Analysis
- **Activity 2.3:** Develop Audio Feature Extraction Module
- **Activity 2.4:** Build Concept Evaluation and Scoring Engine

### **Milestone 3: App.py Development**

- **Activity 3.1:** Develop Streamlit Application
- **Activity 3.2:** Integrate AI Processing Pipeline
- **Activity 3.3:** Implement PDF Report Generation

### **Milestone 4: Frontend Development**

- **Activity 4.1:** Design the User Interface
- **Activity 4.2:** Implement Dynamic Content Rendering
- **Activity 4.3:** Display Evaluation Results
- **Activity 4.4:** Enable Report Download

### **Milestone 5: Deployment**

- **Activity 5.1:** Prepare the Application for Local Deployment
- **Activity 5.2:** Local Testing and Verification
- **Activity 5.3:** Deployment Preparation

### **Milestone 6: Conclusion**

## **Milestone 1: Model Selection and Architecture**

In this milestone, we focus on selecting the appropriate AI models and audio-processing tools for the Voice-Based Concept Understanding Analyser. This involves researching speech-to-text models,

semantic similarity techniques, audio feature extraction methods, and report generation libraries to ensure the application accurately evaluates spoken conceptual explanations.

### Activity 1.1: Install and Configure Required Libraries

Before the application can process audio and evaluate understanding, all required Python libraries must be installed and configured properly.

- **Step 1 —Install Python:** Install Python 3.10 or above from the official Python website.

```
C:\Users\dilee\OneDrive\Desktop\Ihant\Voice Based Concept Understanding Analyser> pip install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: openai-whisper in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 1)) (20250625)
Requirement already satisfied: librosa in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 2)) (0.11.0)
Requirement already satisfied: streamlit in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 3)) (1.58.0)
Requirement already satisfied: reportlab in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 4)) (5.0.0)
Requirement already satisfied: nltk in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 5)) (3.9.4)
Requirement already satisfied: torch in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 6)) (2.12.1)
Requirement already satisfied: sentence-transformers in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 7)) (5.6.0)
Requirement already satisfied: matplotlib in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 8)) (3.11.0)
Requirement already satisfied: soundfile in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 9)) (0.14.0)
Requirement already satisfied: audioread in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 10)) (3.1.0)
Requirement already satisfied: torchvision in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from -r requirements.txt (line 11)) (0.27.1)
Requirement already satisfied: more-iterertools in c:\users\dilee\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from openai-whisper->-r requirements.txt (line 1)) (11.0)
```

- **Step 2 —Create a Virtual Environment:** Create a virtual environment to manage project dependencies separately

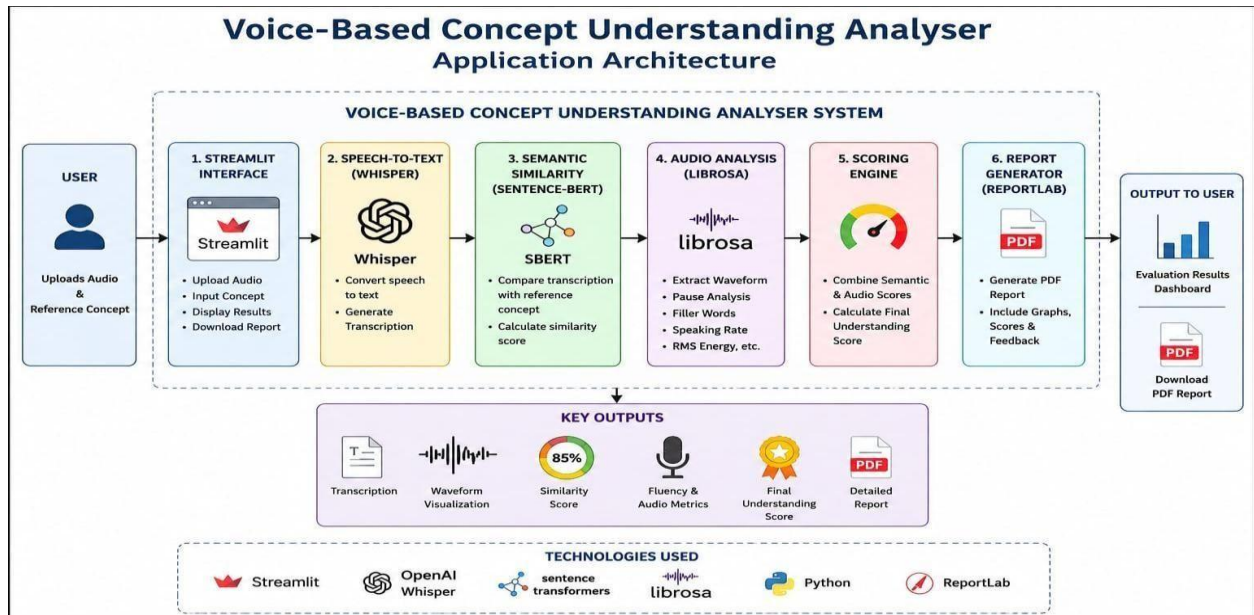
```
import os
import tempfile
import html
import logging

import streamlit as st
import librosa
import librosa.display
import matplotlib.pyplot as plt

from speech_to_text import speech_to_text
from semantic_eval import semantic_similarity
from audio_utils import extract_audio_features, filler_word_ratio
from scoring_engine import evaluate_understanding
from report_generator import generate_pdf_report
```

- **Step 3 — Install Required Libraries:** Install Streamlit, Whisper, Sentence-Transformers, Librosa, Torch, Matplotlib, NLTK, SoundFile, and ReportLab.





- **Step 4 — Verify Installation:** Run the project and confirm that all libraries are imported without errors.

### Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

**Upload Student Audio (WAV)**  
Drag and drop file here

WhatsApp...34 PM.wav 294.3KB

0:00 / 0:23

**Audio Waveform**

Analyze Concept Understanding

Analysis Completed

**Transcribed Explanation**

Machine Learning is a branch of artificial intelligence that allows computers to learn from data and improve their performance without being directly programmed. It works by finding patterns in data and using those patterns to make predictions or decisions. Machine Learning is used in recommendation systems, fraud detection, speech recognition, image classification, healthcare, and self-driving cars.

Semantic Similarity: **86.96**

Filler Word Ratio: **0.0**

Confidence (Energy): **0.0355**

Download PDF Report

**Concept Reference**

Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

**Final Evaluation**

Understanding Score: **86/100**

**Strong Understanding**

- **Step 5 — Run the Application:** Start the Streamlit application using `streamlit run app.py`.

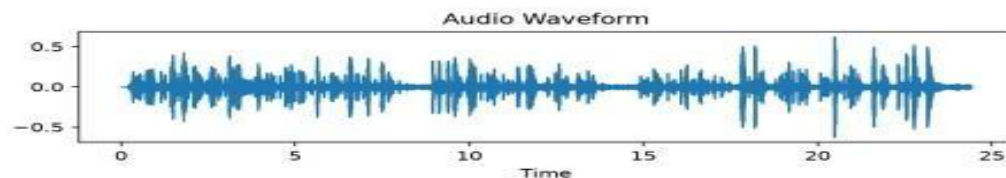
### Reference Concept

Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

### Student Transcription

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### Audio Visualization



### Evaluation Summary

| Metric              | Value                |
|---------------------|----------------------|
| Semantic Similarity | 86.96                |
| Filler Word Ratio   | 0.0                  |
| Pause Ratio         | N/A                  |
| Confidence (Energy) | 0.0355               |
| Final Score         | 86/100               |
| Understanding Level | Strong Understanding |

## Activity 1.2: Research and Select the Appropriate Generative AI Model

The following technologies were researched and selected based on their accuracy, performance, and suitability for spoken concept evaluation.

**Understand the Project Requirements:** Analyze the need for speech transcription, semantic understanding, audio feature extraction, and automated educational assessment.

**Explore AI Model Documentation:** Review the documentation for OpenAI Whisper, Sentence-BERT (all-MiniLM-L6-v2), Librosa, and ReportLab to understand their capabilities and integration process.

**Evaluate Model Performance:** Compare models based on transcription accuracy, semantic similarity performance, computational efficiency, and support for educational applications.

**Select the Optimal Models:** Finalize OpenAI Whisper for speech-to-text transcription, SentenceBERT for semantic similarity evaluation, Librosa for audio signal analysis, and ReportLab for PDF report generation.

## Activity 1.3: Define the Architecture of the Application

**Design the System Architecture:** Create a modular architecture consisting of the Streamlit frontend, AI processing modules, evaluation engine, and PDF report generator.

**Define Frontend Workflow:** Design an interface that allows users to upload audio, view waveform visualization, analyze conceptual understanding, and download evaluation reports.

**Develop Backend Modules:** Implement modules for speech transcription, semantic similarity calculation, audio feature extraction, scoring engine, and report generation.

**Integrate AI Components:** Connect OpenAI Whisper, Sentence-BERT, Librosa, Matplotlib, and ReportLab to build a complete AI-powered evaluation pipeline.

### Activity 1.4: Set Up the Development Environment

Set up the Python environment and project structure required to run the Voice Based Concept Understanding Analyser application.

- **Install Python and Pip:** Ensure Python 3.10+ is installed along with pip for package management.
- **Create a Virtual Environment:** Create a separate virtual environment to manage project dependencies safely.

```
python -m venv venv
```

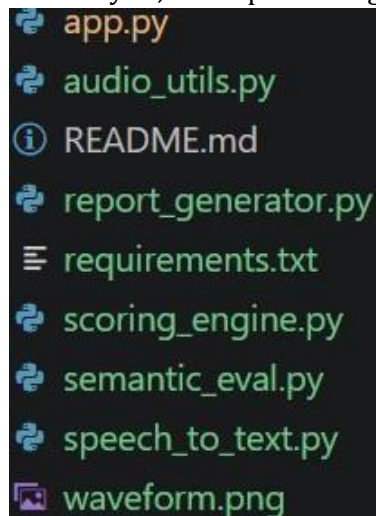
- **Activate the Virtual Environment:**

```
venv\Scripts\activate
```

- 

```
pip install -r requirements.txt
```

- **Set Up Application Structure:** Organize the project files into separate modules for transcription, semantic analysis, audio processing, scoring, and report generation.



```
app.py
audio_utils.py
README.md
report_generator.py
requirements.txt
scoring_engine.py
semantic_eval.py
speech_to_text.py
waveform.png
```



## **Milestone 2: Core Functionalities Development**

### **Activity 2.1 Develop Speech-to-Text Transcription Module**

- Implement speech transcription using OpenAI Whisper.
- Accept uploaded WAV audio files.
- Convert speech into accurate text.
- Handle different accents and background noise.

### **Activity 2.2: Implement Semantic Similarity Analysis**

- Compare the student's explanation with the reference concept.
- Generate semantic similarity using Sentence-BERT.
- Measure conceptual understanding accurately.
- Produce similarity score for evaluation.

### **Activity 2.3: Develop Audio Feature Extraction Module**

- Extract waveform using Librosa.
- Calculate filler word ratio.
- Measure pause ratio and RMS energy.
- Analyze communication confidence.

### **Activity 2.4: Build Concept Evaluation and Scoring Engine**

- Combine transcription, semantic similarity, and audio features.
- Calculate the final understanding score.
- Classify performance as Strong, Moderate, or Poor Understanding.
- Generate structured evaluation results and PDF report.

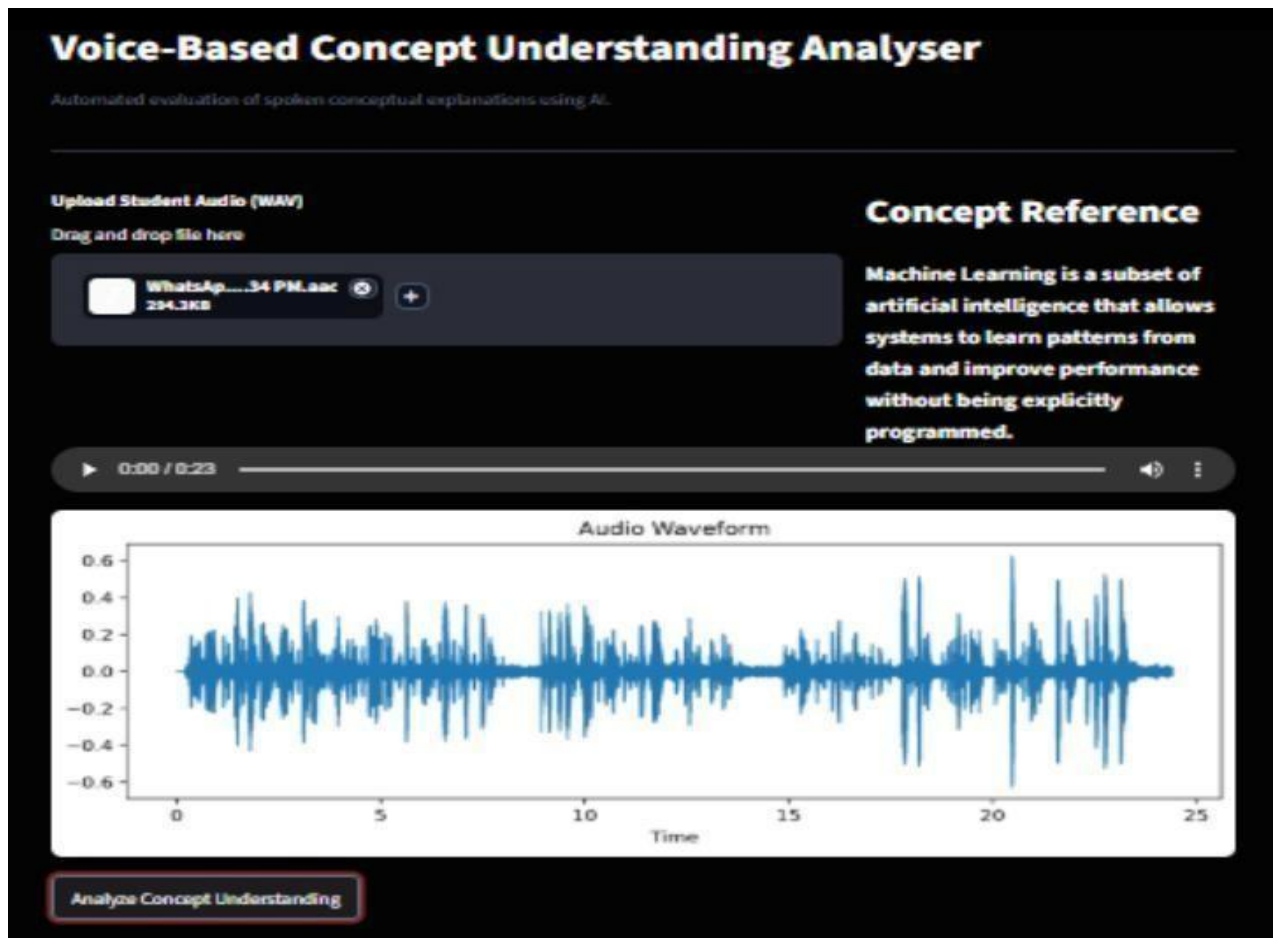
## **Milestone 3: App.py Development**

Milestone 3 focuses on developing the main application file `app.py`, which connects all core modules of the Voice-Based Concept Understanding Analyser. This file handles the Streamlit interface, audio

upload, reference concept input, transcription, semantic similarity analysis, audio feature extraction, scoring, waveform display, and PDF report generation. Activity 3.1: Writing the Main Application Logic in app.py

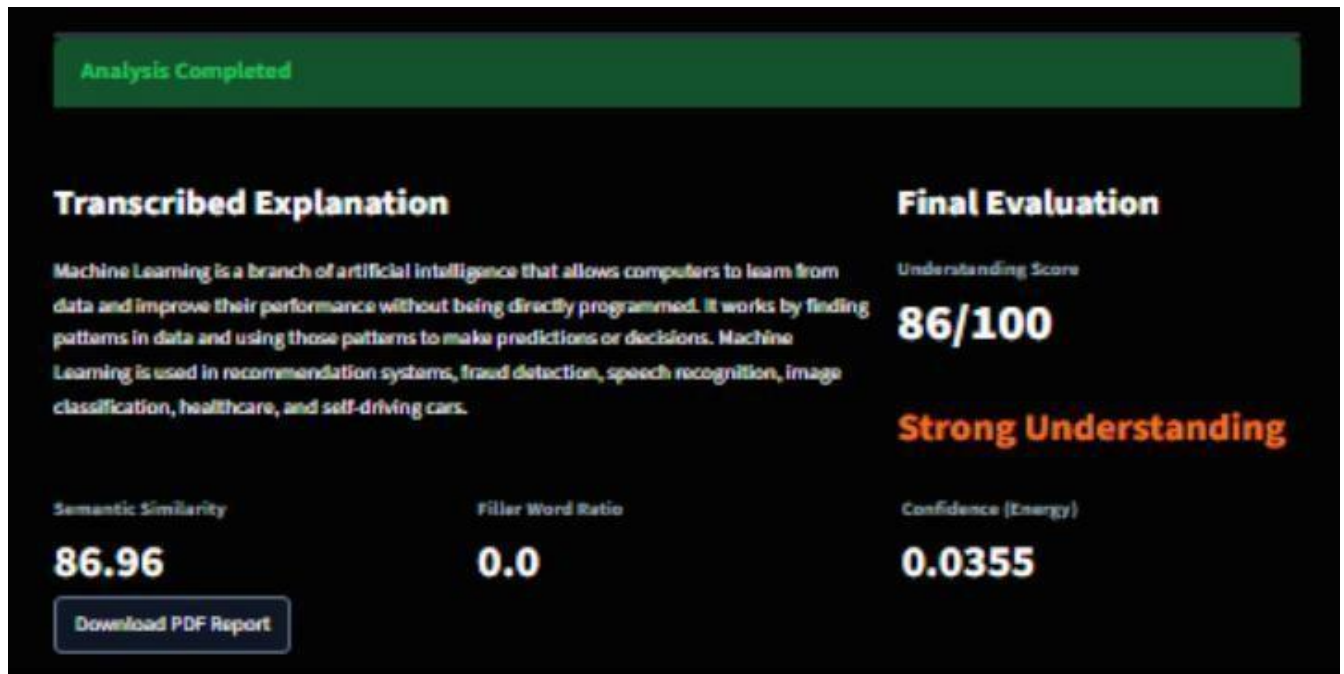
## Activity 3.1: Develop Streamlit Application

- The main Streamlit application is developed in app.py. It provides the user interface where users
- can upload audio, view the reference concept, listen to the uploaded recording, and start the
- analysis process.
- Create the application title and description.
- Add audio file upload option.
- Display the reference concept section.
- Show uploaded audio playback.
- Provide the Analyze Concept Understanding button.



## Activity 3.2: Integrate AI Processing Pipeline

- The AI pipeline is integrated inside app.py by connecting transcription, semantic similarity, audio feature extraction, and scoring modules.
- Convert uploaded audio into text using the speech-to-text module.
- Compare transcription with the reference concept using semantic similarity.
- Extract audio features such as filler word ratio, pause ratio, RMS energy, and waveform.
- Calculate final understanding score using the scoring engine.
- Display analysis status and output values on the dashboard.



## Activity 3.3: Implement PDF Report Generation

- The PDF report generation feature is connected to the main application so users can download their evaluation summary.
- Generate a structured PDF report after analysis.
- Include reference concept, student transcription, waveform visualization, evaluation metrics, final score, and understanding level.
- Provide a Download PDF Report button inside the Streamlit interface.
- Save reports inside the reports folder for future reference.

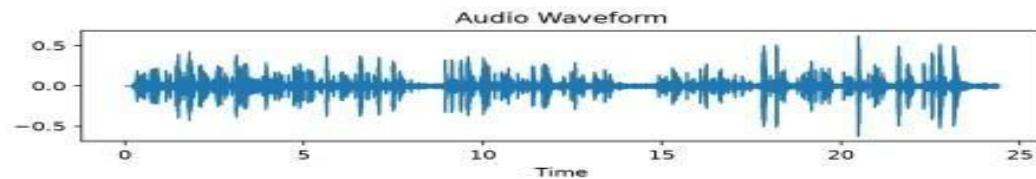
## Reference Concept

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## Student Transcription

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## Audio Visualization



## Evaluation Summary

| Metric              | Value                |
|---------------------|----------------------|
| Semantic Similarity | 86.96                |
| Filler Word Ratio   | 0.0                  |
| Pause Ratio         | N/A                  |
| Confidence (Energy) | 0.0355               |
| Final Score         | 86/100               |
| Understanding Level | Strong Understanding |

## Verification and Output:

After integrating all modules successfully, the application performs end-to-end processing whenever a user uploads an audio explanation. The uploaded speech is transcribed, evaluated using semantic similarity analysis, and assessed for fluency through audio feature extraction. The scoring engine combines these results to generate the final concept understanding score. The application displays the transcription, waveform, evaluation metrics, and qualitative feedback in the Streamlit interface. Finally, a detailed PDF report containing all analysis results is generated and made available for download, providing users with a complete summary of their performance.

## Milestone 4: Frontend Development

### Activity 4.1: Designing and Developing the User Interface

Milestone 4 focuses on designing a user-friendly Streamlit interface for the Voice-Based Concept Understanding Analyser. The frontend allows users to upload audio, view the reference concept, analyze speech, display waveform visualization, show evaluation metrics, and download the final PDF report.

#### Activity 4.1: Design the User Interface

- Create the Streamlit page title: Voice-Based Concept Understanding Analyser.
- Add audio upload option for .wav files.
- Display the reference concept section.
- Add audio playback for uploaded files.
- Add the Analyze Concept Understanding button.

#### Activity 4.2: Implement Dynamic Content Rendering

- Show waveform automatically after audio upload.
- Display Analysis Completed message after processing.
- Render transcription, scores, and feedback dynamically.
- Update the dashboard based on the uploaded audio file.

#### Activity 4.3: Display Evaluation Results

- Display student transcription.
- Show semantic similarity score.
- Show filler word ratio, pause ratio, and confidence value.
- Display final score and understanding level.
- Highlight output as Strong Understanding, Moderate Understanding, or Poor Understanding.

#### Activity 4.4: Enable Report Download

- Generate a PDF report using ReportLab.
- Include reference concept, transcription, waveform, metrics, final score, and feedback.
- Provide Download PDF Report button.
- Allow users to save the report for review.



## **Frontend Testing and Validation:**

The frontend was thoroughly tested to ensure smooth interaction between the user interface and backend processing modules. All interface components, including audio upload, waveform visualization, transcription display, semantic analysis, scoring, and report generation, functioned correctly without errors. Dynamic content updates were verified after each analysis, ensuring users received real-time feedback.

The application was also tested across multiple audio samples to validate responsiveness, accuracy, and usability. The PDF report generation feature successfully exported evaluation results, making the system suitable for academic assessments, concept evaluation, interview preparation, and spoken learning analysis. Overall, the frontend provides an intuitive, responsive, and interactive experience while effectively presenting AI-generated insights and performance metrics.

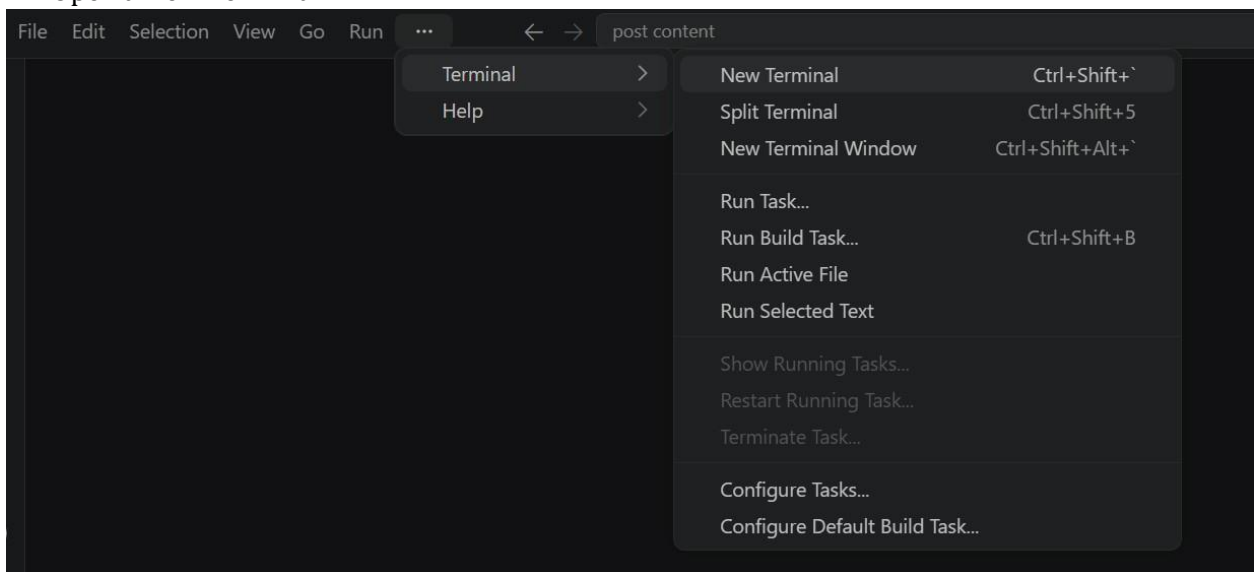
## **Milestone 5: Deployment**

Milestone 5 focuses on deploying the Voice-Based Concept Understanding Analyser for local testing and public access. This milestone ensures that all required dependencies are installed, the Streamlit

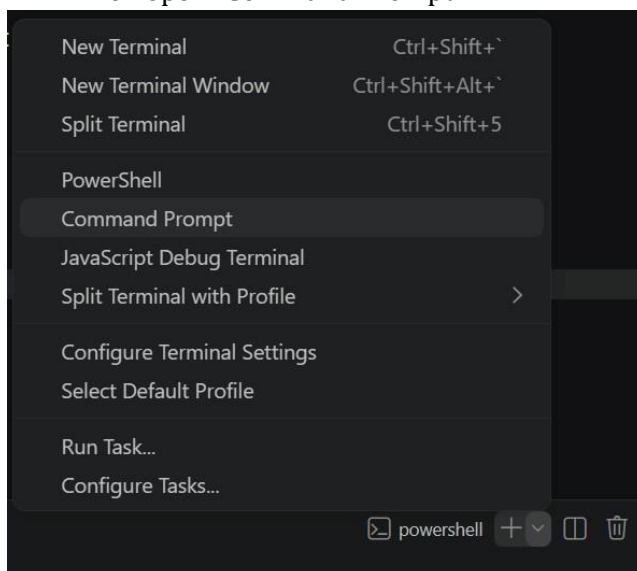
application runs successfully, and the project can be securely shared using Ngrok for demonstration and evaluation.

## Activity 5.1: Preparing the Application for Local Deployment

- Set Up the Development Environment
- Open the project in Visual Studio Code.
- Create and activate a Python virtual environment.
- Install all required libraries using the requirements.txt file.
- Verify that all project modules and dependencies are installed successfully.
- Configure the project structure and required folders before execution. **Set Up a Virtual Environment:**
- Open a New Terminal



- Then open “Command Prompt”



- Create and activate a virtual environment, then install all required dependencies from requirements.

```
D:\projects>python -m venv myenv  
D:\projects>"d:\projects\venv\Scripts\activate"  
(myenv) D:\projects>pip install -r requirements.txt
```

### Activity 5.2: Local Testing and Verification

- Launch the Streamlit application using the terminal.
- Open the local Streamlit URL in the browser.
- Upload a sample audio file for testing.
- Verify speech transcription, semantic similarity analysis, waveform visualization, scoring, and PDF report generation.
- Confirm that all application modules execute successfully without errors.

```
(myenv) D:\projects>python app.py
```

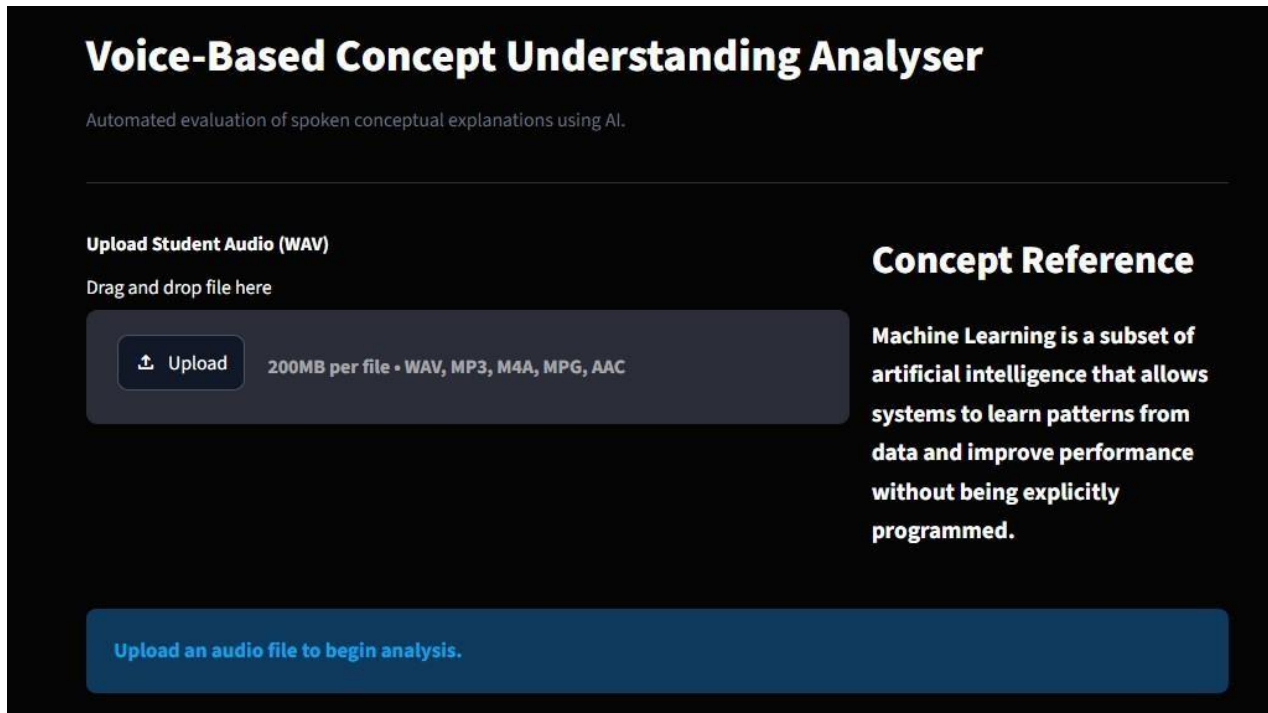
```
PS C:\Users\dilee\OneDrive\Desktop\Than1\Voice Based Concept Understanding Analyser> python -m streamlit run app.py  
2026-06-27 12:11:16.935 Uvicorn server started on 0.0.0.0:8501  
  
You can now view your Streamlit app in your browser.  
  
Local URL: http://localhost:8501  
Network URL: http://192.168.31.66:8501
```

### Activity 5.3: Deployment Preparation

- Verified that all required Python dependencies were installed successfully.
- Ensured the Streamlit application was configured correctly for execution.
- Tested the application in the local development environment.
- Verified that speech transcription, semantic analysis, audio feature extraction, scoring, waveform visualization, and PDF report generation worked correctly.

- Prepared the project for future deployment on cloud platforms such as Streamlit Community Cloud, Render, or Ngrok.

```
D:\projects>pip install pyngrok
```



### Verification:

The application was successfully executed in the local environment. All core modules functioned as expected, including speech-to-text transcription, semantic similarity analysis, audio feature extraction, intelligent scoring, waveform visualization, and PDF report generation. The project is deployment-ready and can be hosted on a public platform in the future if required.

## Exploring the Website's Web Pages:

### Home / Generator Page:

## Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

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**Upload Student Audio (WAV)**

Drag and drop file here

 Upload

200MB per file • WAV, MP3, M4A, MPG, AAC

**Concept Reference**

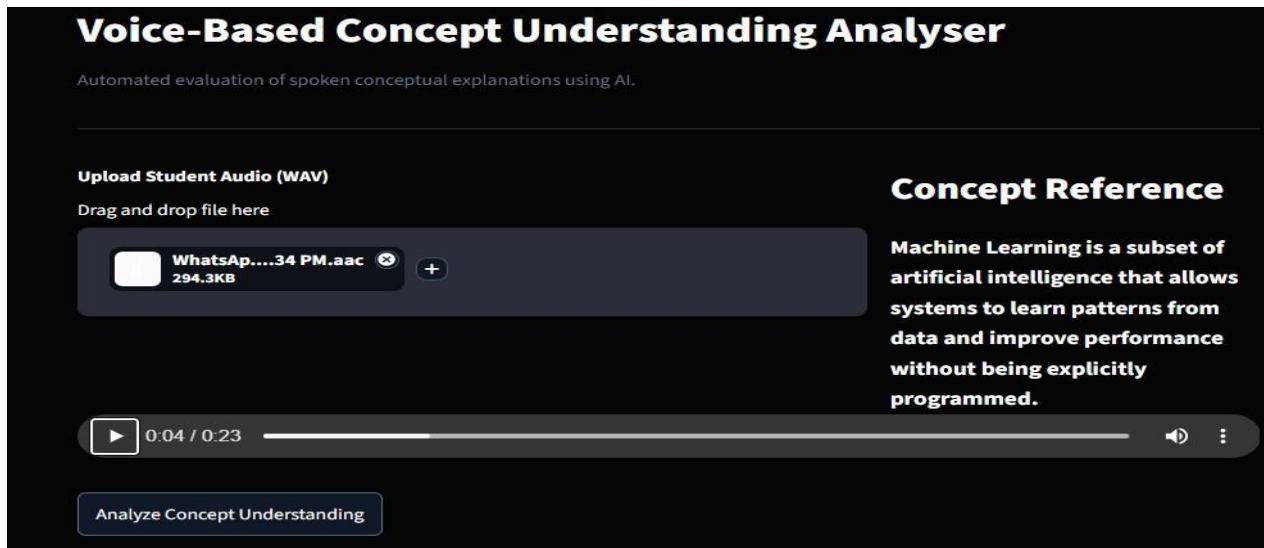
**Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.**

Upload an audio file to begin analysis.

**Description:** The Home Page serves as the main interface of the Voice-Based Concept

**Understanding Analyser:** Users can upload an audio recording, enter the reference concept, and start the analysis process. The interface is designed using Streamlit to provide a simple, responsive, and user-friendly experience.

## Input Section:



**Voice-Based Concept Understanding Analyser**  
Automated evaluation of spoken conceptual explanations using AI.

**Upload Student Audio (WAV)**  
Drag and drop file here

WhatsApp....34 PM.aac 294.3KB

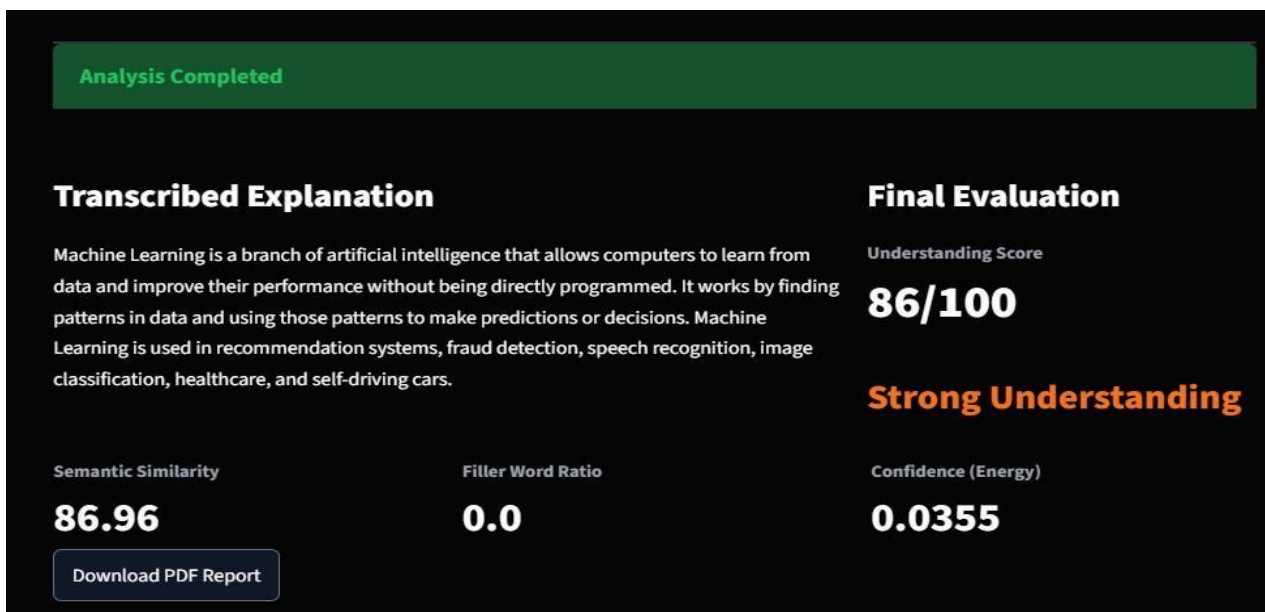
**Concept Reference**  
Machine Learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve performance without being explicitly programmed.

0:04 / 0:23

Analyze Concept Understanding

**Description:** The Input Section allows users to upload a voice recording in WAV format and enter the expected concept or topic. Once the required inputs are provided, the user can initiate the analysis by clicking the Analyze Concept Understanding button.

## Results Section:



**Analysis Completed**

**Transcribed Explanation**  
Machine Learning is a branch of artificial intelligence that allows computers to learn from data and improve their performance without being directly programmed. It works by finding patterns in data and using those patterns to make predictions or decisions. Machine Learning is used in recommendation systems, fraud detection, speech recognition, image classification, healthcare, and self-driving cars.

**Final Evaluation**  
Understanding Score  
**86/100**  
**Strong Understanding**

Semantic Similarity: **86.96**  
Filler Word Ratio: **0.0**  
Confidence (Energy): **0.0355**

Download PDF Report

**Description:** The Results Section displays the generated speech transcription, semantic similarity score, audio feature analysis, waveform visualization, filler word statistics, pause analysis, confidence



score, and the final understanding level. These results help users evaluate both conceptual understanding and communication skills.

## PDF Report Section:

### Reference Concept

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### Audio Visualization



### Evaluation Summary

| Metric              | Value                |
|---------------------|----------------------|
| Semantic Similarity | 86.96                |
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| Pause Ratio         | N/A                  |
| Confidence (Energy) | 0.0355               |
| Final Score         | 86/100               |
| Understanding Level | Strong Understanding |

**Description:** After the evaluation is completed, the application generates a detailed PDF report containing the transcription, semantic similarity score, waveform image, speech metrics, final evaluation score, and personalized feedback. Users can download the report for future reference or academic assessment.

## Conclusion:

The Voice-Based Concept Understanding Analyser (VBCUA) successfully integrates speech-to-text transcription, semantic similarity analysis, audio feature extraction, intelligent scoring, and automated PDF report generation into a single AI-powered educational platform. Built using Streamlit, OpenAI Whisper, Sentence-BERT, Librosa, Matplotlib, and ReportLab, the application provides accurate evaluation of conceptual understanding and spoken communication. The project demonstrates the effective application of Artificial Intelligence, Natural Language Processing, and Audio Signal Processing to create an interactive assessment tool for students, educators, trainers, and researchers. It can be further enhanced by supporting multiple languages, real-time speech analysis, cloud deployment, and personalized learning recommendations.